Research Square

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EXPLORE: learning interpretable rules for patient-level prediction

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Abstract

Objective

We investigate whether a trade-off occurs between predictive performance and model interpretability in real-world health care data and illustrate how to develop *clinically optimal* decision rules by learning under constraints with the Exhaustive Procedure for Logic-Rule Extraction (EXPLORE) algorithm.

Methods

We enhanced EXPLORE's scalability to enable its use with real-world datasets and developed an R package that generates simple decision rules. We compared EXPLORE's performance to 7 state-of-the-art model algorithms across 5 prediction tasks using data from the Dutch Integrated Primary Care Information (IPCI) database. Additionally, we characterized EXPLORE's space of near-optimal models (i.e. Rashomon set) and conducted experiments on incorporating domain knowledge and improving existing models.

Results

The prediction models developed using LASSO, RandomForest, and XGBoost consistently performed best in terms of AUROC, followed by DecisionTree and EXPLORE. However, the decision rules generated by EXPLORE are much simpler (at most 5 predictors) than the aforementioned. GOSDT-G, IHT, and RIPPER performed worse. Moreover, we demonstrated that EXPLORE's Rashomon set is very large (1,381 – 20,320 models) with a large variability in both the generalizability and model diversity. We then showed there is a potential to find more clinically optimal decision rules using EXPLORE by incorporating domain knowledge (age/sex and task-specific features) or improving existing models (the $CHADS_2$ score).

Conclusions

Our study shows that more complex models generally outperform simpler ones, confirming the expected interpretability-performance trade-off, although it varies in strength across prediction tasks. EXPLORE's ability to learn under constraints is valuable for generating clinically optimal decision rules.

1. BACKGROUND

Rule-based methods have been among the most popular classification methods in machine learning since the 1960s (1). As predictions of these models are the result of a series of relatively simple computations, small rule-based models are often considered to be interpretable (2). Interpretable models allow users to understand, verify, and improve model output (3), which is often demanded when the cost of misclassification is high (e.g. in health care). In addition, these models may be easier to implement as they can be used without system integration and face less technical challenges during deployment due to their smaller size. Moreover, rule-based methods show close resemblance to existing clinical workflows such as knowledge-based expert systems and clinical pathways. Despite these advantages, rule-based methods are less frequently used to develop clinical prediction models compared to other methods like regression analysis (4).

In this paper we aim to develop *clinically optimal* decision rules using electronic health record (EHR) data. The concept of *optimality* in clinical prediction modelling goes beyond the traditional mathematical definition of a feasible solution

that minimizes (or maximizes) a given objective function. Model performance is important, but for a model to be successful in practice more is needed. First, a clinically optimal model is interpretable, i.e., as simple as possible in both model form and size as well as the features included. A model (partially) aligning with prior knowledge might be better generalizable and more readily adopted in practice. Second, a clinically optimal model might need to satisfy certain constraints (e.g. a minimum level of specificity). Third, a clinically optimal model is stable when updated. Similar features are included in the model and predictions for similar examples are similar, all while achieving performance that is (close to) the 'best' possible for the given prediction task.

The Exhaustive Procedure for Logic-Rule Extraction (EXPLORE) algorithm (5) has the necessary features to develop such clinically optimal decision rules: it produces small decision rules, has optimality guarantees, and allows the user to specify global model constraints. The contributions of this paper are as follows:

- We improve EXPLORE (5) in terms of scalability, making it accessible and executable for real-world sized datasets. We demonstrate this by developing decision rules for five prediction tasks (~ 50 features, ~ 4,000-200,000 observations) using Dutch general practitioner (GP) data.
- We investigate whether a trade-off occurs between predictive performance and model interpretability in real-world health care data by comparing EXPLORE's performance to seven other commonly used state-of-the-art modelling methods in a large-scale empirical evaluation.
- We illustrate how to develop clinically optimal decision rules by learning under constraints with EXPLORE.

2. RELATED WORK

2.1. Optimal rule-based model algorithms

Rule-based methods are a popular choice to develop interpretable models as they are human understandable (consist of a collection of statements connected using "IF-THEN", "OR", and "AND") and allow to model categorical data with non-linear relationships. Examples include decision trees, decisions lists (i.e. ordered sets of rules or one-sided decision trees), and decision sets (i.e. unordered sets of rules). Historically, greedy approaches have often been used to grow decision trees (e.g. CART, C4.5) and decision lists/sets (e.g. RIPPER, C4.5R), as decision tree optimization is known to be an NP-complete problem (6). These approaches split the optimization problem into steps and search for the optimal solution at every split. Although this makes their computation more tractable, the disadvantage is that there are no optimality guarantees for their final performance. In case of bad performance, it is therefore unclear whether this is due to model choice or poor optimization.

In recent years several methods have been proposed to grow *optimal decision trees*. For example, DL8.5 (7) combines dynamic programming on the space of the decision trees and branch-and-bound optimization, BinOCT (8) uses a novel binary linear program formulation for optimization, and GOSDT (9) uses dynamic programming with custom bounds. However, DL8.5 and BinOCT loose optimality when using bucketization to handle continuous features. Although less work focuses on *optimal decision rules*, some methods with optimization guarantees have been proposed: EXPLORE (5) uses a branch-and-bound approach for efficient exhaustive search, IDS (10) relies on submodular optimization, CG (11) uses a column generation algorithm, and LIBRE (12) creates an ensemble of weak learners.

The exhaustive search performed by EXPLORE offers several advantages over other methods as it allows users to specify global constraints on the model (e.g. model size, included features) or its performance (e.g. minimum sensitivity and specificity). Earlier work investigating the performance of EXPLORE on UCI datasets has shown promising results (5), but is limited as the studied prediction tasks do not match the complexity of real-world data.

2.2. Interpretability-performance trade-off

Ideally, we want models to produce the best possible predictions, while being simple enough to be understood by users. The interpretability-performance trade-off refers to the general assumption that increased model flexibility contributes to a better model fit (increasing performance) at the cost of additional complexity (reducing interpretability). Rudin (13) raised important awareness that this trade-off might be a myth and stressed the importance of using interpretable models especially for high-stake decisions. She argued that especially for structured data with meaningful features increased interpretability does not necessarily lead to a loss of performance.

There is a growing body of research questioning the extent to which this trade-off occurs (13–15). Various studies have shown examples where complex models do not outperform simpler models (16–20). Furthermore, experiments in various domains show there might be multiple equally-good performing models, leading some to suggest that interpretable models are likely to exist among them (21). The extent of the trade-off might ultimately depend on the data characteristics (18) and used metrics (14). Also for medical data, it is often assumed that this trade-off exists (22–24).

3. METHODS

3.1. EXPLORE R package

We developed the EXPLORE R package (https://github.com/mi-erasmusmc/Explore), which implements the EXPLORE algorithm (5) that generates optimal decision rules of specified length in disjunctive normal form (DNF). A formula in DNF is a disjunction (OR, $\vee$) of terms, where each term is a conjunction (AND, $\wedge$) of literals, i.e. feature-operator-value triples (e.g. age > 40). The rules are systematically generated using three nested loops that evaluate the next term tuple (i.e. ordered list of term sizes), feature-operator pair, and threshold value, respectively (see Fig. 1). For example, when creating a rule with maximum length 3 for a dataset with 10 binary predictors EXPLORE generates 5540 candidate rules. For details on the original algorithm we refer to the original publication (5).

We made two main contributions to improve EXPLORE's scalability. First, we further optimized the generation of decision rules for binary data. In particular, we show that it is possible to simplify rules including a term of size 1 to a rule of shorter length using the redundancy law of Boolean algebra (see Appendix A.1). This reduces the number of feature-operator pairs that need to be generated, leading to a reduction in the number of fully instantiated rules of 7.0% for rule length 3 to 20.5% for rule length 5. Second, we parallelized EXPLORE using the RcppParallel package. Using the improved version of EXPLORE we reduced average computation time by 92.2% for rule length 4 (see Appendix A.2).

3.2. Dataset

We used real-world health data from the Dutch Integrated Primary Care Information (IPCI) database (25). This database contains longitudinal, routinely collected data from computer-based patient records of around 350 GP practices throughout the Netherlands. In total, the database currently (January 1, 2024) contains 2.87 million patient records with a median follow-up duration of 4.7 years. The number of active patients is 1.3 million, which comprises about 8% of the Dutch population (~ 17 million). Data include patient demographics, information about contacts with GPs, symptoms, diagnoses, laboratory and clinical measurements, prescriptions, and information on use of secondary care. The IPCI database has been mapped to the Observational Medical Outcomes Partnership Common Data Model (OMOP CDM), which enables standardized extraction and analysis of health care data. This study was approved by the IPCI Governance Board (number 3/2023).

3.3. Prediction tasks

We selected five prediction tasks for evaluation, for their definition see Table 1. These tasks represent a range of different target populations (generic and disease specific), time-at-risks (from 1 month up to 5 years), and outcome

rates. Furthermore, these prediction tasks have a clear clinical use case and have been more commonly investigated in the literature: hospital readmission (26, 27), end-of-life care (28, 29), asthma exacerbations (30, 31), mortality in chronic obstructive pulmonary disease (COPD) (32), and cardiovascular disease in type-2 diabetes mellitus (T2DM) (33). We restricted all target populations to adults (age > = 18) and include events in the last five years before database end minus the time-at-risk period (e.g. for hospital readmission, cohort entry date after December 1, 2018).

Table 1
Specification of prediction tasks. Generic prediction question has the form (34): "among *target population*, which patients will develop *outcome of interest* during *time-at-risk period*?"

Task	Target population	Outcome of interest	Time-at-risk period
Hospital readmission	Adult patients discharged from hospital	Hospital admission	1 month
End-of-life care	GP visit of older patients (60+) with a prior diagnosis of heart failure, COPD, or cancer	End-of-life conversation	6 months
Asthma exacerbations	Adult patients with new asthma diagnosis receiving medication	Exacerbation (as defined by 3–30 days of systemic corticosteroids)	2 years
Mortality in COPD	Patients with new COPD diagnosis (40+)	All-cause mortality	2 years
Cardiovascular disease in T2DM	Adult patients with new T2DM diagnosis	Heart failure or stroke	5 years

3.4. Model development

We used the PatientLevelPrediction R package (34) to develop prediction models. This tool provides a streamlined pipeline to develop clinical prediction models in line with current best practices using OMOP CDM data. The tool extracts the necessary information from the database based on the specified prediction task, splits the data into a training (75%) and test set (25%), constructs and pre-processes the covariates (see Section 3.5), fits models using different modelling algorithms on the training set (see Section 3.6), and performs internal validation of the model on the test set (see Section 3.7).

3.5. Candidate predictors

The predictors included for model development were patient demographics (i.e. age and sex) and binary indicators for the presence of 47 clinical events in the 365 days prior to the index date. The set of clinical events - including common medical conditions and drug exposures - is based on the constrained set of clinically meaningful predictors identified by Reps, Wong (35) to achieve minimal performance loss across a variety of prediction tasks (when compared to a large set of predictors). We removed predictors that occurred in less than 0.1 percent of the target population.

3.6. Model algorithms

In addition to EXPLORE, we investigated 7 machine learning algorithms that are commonly used in clinical prediction modelling or rule-based learning: DecisionTree (36), GOSDT-G (37), Iterative Hard Thresholding (IHT) (38), LASSO (39), RandomForest (40), RIPPER (41), and XGBoost (42). We used 3-fold cross validation to optimize the hyperparameters, including a range of values for each model algorithm to allow for varying levels of model complexity. For details on used libraries and hyperparameters we refer to Appendix B.

3.7. Model evaluation

Predictive performance

The area under the receiver operating characteristic (AUROC) curve was used to evaluate the ability of the model to distinguish between the classes. Some of the included model algorithms predict classes instead of probabilities (DecisionTree, EXPLORE, GOSDT-G, RIPPER). These models do not allow to vary the risk threshold and hence have a fixed number of true/false positives. EXPLORE's ability to learn under constraints opens possibilities to evaluate its performance in a similar way as models predicting probabilities (despite it predicting classes). Details on how to create the AUROC curve for EXPLORE by developing models for different constraints on sensitivity and specificity can be found in Appendix C.

Interpretability

Model size was used as a proxy for model interpretability (17, 43). For EXPLORE and RIPPER model size is measured by the total length of rules. In the case of tree-based algorithms like DecisionTree, GOSDT-G, RandomForest and XGBoost, we consider the total number of nodes in the tree(s). For LASSO and IHT, we use the number of non-zero coefficients in the model. As various proxies can be chosen for each model algorithm, we performed a sensitivity analysis by counting the number of unique features used per model. While the first metric captures to the number of computations required for model application and aims to represent the complexity of the model's decision-making process, the second metric represents the amount of information a clinician must collect (e.g. by asking questions) to use the model.

3.8. Experimental setup

We developed a total of 40 models (8 model algorithms x 5 prediction tasks) to evaluate the performance of EXPLORE and more generally investigate the existence of the interpretability-performance trade-off. In addition, we conducted experiments to illustrate how EXPLORE's learning under constraints can be used to develop clinically optimal decision rules.

Quantifying the search space of good candidate models

The phenomenon that different prediction models relying on different features can perform (almost) equally well is referred to as the Rashomon effect (44). The Rashomon set is defined as the collection of near-optimal models, and its size can be measured by comparing it to the total number of possible models, resulting in the Rashomon ratio (21). Although the Rashomon set and ratio are difficult to compute (exactly) for most model classes (due to their hypothesis space being infinite), these can easily be computed for EXPLORE using its ability to learn under constraints. We constructed the Rashomon set for each prediction task by setting a global model constraint at 99% of the maximum balanced accuracy (i.e. performance of the best model). We further examined the Rashomon set in terms of generalizability (i.e. whether training performance transfers to the test set) and model diversity (i.e. frequency of included features and variation using the model stability metric of Nogueira, Sechidis (45))).

Incorporating domain knowledge and improving existing models

Next, we investigated whether we can enforce constraints to include clinically relevant features for the prediction tasks without losing performance. Two cases were considered: a) including age/sex and b) two task-specific features selected based on clinical knowledge, the remaining predictors were in both cases selected by EXPLORE. Furthermore, we investigated if we can improve the performance of an existing model, the $CHADS_2$ score (46), which estimates the risk of stroke within 1 year for patients with atrial fibrillation. The original $CHADS_2$ score includes the following covariates: congestive heart failure history, hypertension history, age, diabetes mellitus history, stroke or transient ischemic attack (TIA) symptoms. We compared the prediction of the original model score with two updated versions: a) we retrained the model using EXPLORE with the same set of covariates, and b) allowed EXPLORE to add 4 additional covariates on top of the $CHADS_2$ score (included as numerical feature).

The full code to run our experiments (including all cohort definitions) is available on GitHub: https://github.com/mi-erasmusmc/ExploreExperiments. Experiments can be repeated on (other) databases mapped to the OMOP CDM.

4. RESULTS

4.1. Baseline characteristics

Table 2 presents a summary of the five prediction tasks. As shown the experiments include a variety of tasks with different target population sizes (4,002–230,952) and outcome rates (0.9% – 29.1%), demographics vary across tasks as expected. An overview of all clinical characteristics is included in Appendix D.

Table 2
Characteristics of the prediction tasks.

	Hospital readmission	End-of-life care	Asthma exacerbations	Mortality in COPD	Cardiovascular disease in T2DM
Target population size	75,375	230,952	4,002	4,094	6,634
Age, mean ± SD	63.6 ± 17.9	71.6 ± 8.3	48.3 ± 17.7	67.1 ± 11.1	60.2 ± 12.4
Birth sex, % female	51.9	52.8	61.2	47.5	44.5
Number of outcomes (% of population)	7,512 (10.0)	2,171 (0.9)	1,164 (29.1)	495 (12.1)	458 (1.6)

4.2. Performance comparison of model algorithms

The maximum AUROC value across prediction tasks and model algorithms ranges from 0.65 (for Hospital Readmission) to 0.77 (for Mortality in COPD) on the test set. Figure 2 shows the difference in the performance of the different model algorithms based on discrimination. AUROC curves and values are included in Appendix D. Across the prediction tasks, LASSO, RandomForest, and XGBoost resulted in models with the best predictive performance. These models show superior performance to the other algorithms, but there is no clear winner among them. The predictive performance in terms of AUROC value for EXPLORE ranged between 0.61 and 0.71 across the prediction tasks. GOSDT-G, and IHT are sometimes close in performance to EXPLORE, but perform worse on average and show a larger variability between tasks. The performance of GOSDT-G may be suboptimal due to the range of selected hyperparameters, which were chosen to remain computationally feasible for our size dataset. DecisionTree performed on average slightly better in terms of AUROC than EXPLORE, whereas RIPPER showed consistently poor performance.

4.3. Model complexity and its relation to predictive performance

We also investigated the relation with model interpretability. The most interpretable models (model size across tasks indicated between brackets) were found by EXPLORE (4–5), GOSDT-G (2–3), IHT (4–9), and RIPPER (5–21). DecisionTree (21–31), LASSO (24–39), XGBoost (259–3935), and RandomForest (628-35028) resulted in more complex models. For a complete overview of model sizes we refer to Appendix D.

Figure 3 shows the relation between interpretability (defined here as the inverse of model size) and performance. A vertical (or horizontal) line indicates no interpretability-performance trade-off. We observe a strong variability in the performance of simple models, while the more complex models show a performance increase that is accompanied by a large reduction in interpretability. These findings confirm the expected trade-off between model performance and

interpretability. However, this observed trade-off varies across prediction tasks. For example, the observed trade-off is stronger for Cardiovascular disease in T2DM than for Asthma exacerbations. In the latter case, a more interpretable model can be chosen with less reduction in performance. Overall, trade-off patterns were found to be similar in the training and test set. Similar results were observed for the sensitivity analysis counting the number of unique features used per model (Appendix D).

4.4. Quantifying the search space of good candidate models

The number of models with less than 1% deviation from the best performance ranges from 153–675 for maximum rule length 4 and 1,381 – 20,320 for maximum rule length 5. Hence, there are many good candidate models for each prediction task. Figure 4a shows the performance of all models in the Rashomon set with maximum rule length 5 compared to the best model (i.e. the model that would be selected during standard model development). For some tasks the best model in the training set is among the best performing on the test set (e.g. End-of-life care and Hospital Readmission), but for the other tasks there exist models in the Rashomon set with much better generalization performance. Figure 4b shows there is a large number of features that are included in one of the models in the Rashomon set and only a handful that are frequently occurring (e.g. age is always included for End-of-life care, Mortality in COPD, and Cardiovascular disease in T2DM). Most features (84–90%) are occurring in less than 10% of the unique feature sets and only 4–6% of the features in more than 75%. According to the stability metric of Nogueira, Sechidis (45), there is most stability in the Rashomon set of Mortality in COPD (0.58) and least for End-of-life care (0.36).

4.5. Incorporating domain knowledge and improving existing models

The previous results suggest there is space to search for models with higher face value and good overall performance, which here we demonstrate with two experiments based on clinical input. Figure 5a shows we find rules with very similar performance for age/sex and the task-specific features: there is less than 0.020 and 0.005 deviation in balanced accuracy and AUROC points, respectively (for both training and test sets). The constraints further lead to significant reductions in computation time (depending on the task and features). Figure 5b shows the performance improvements when using EXPLORE to update the existing $CHADS_2$ model. We identified 88,803 patients in IPCI with newly diagnosed atrial fibrillation of whom 1,422 experienced stroke in the next year (1.6%). The original $CHADS_2$ score has an AUROC value of 0.79 in the test set, which already demonstrates good discriminative model performance. However, when retraining using EXPLORE with $CHADS_2$ as feature, we obtain the best performing model with an AUROC of 0.83.

5. DISCUSSION

5.1. Interpretability-performance trade-off

We have studied the predictive performance and interpretability of eight model algorithms across five prediction tasks in the Dutch GP database IPCI. LASSO, RandomForest, and XGBoost consistently performed best in terms of AUROC, followed by DecisionTree and EXPLORE. As the decision rules generated by EXPLORE are much simpler (at most 5 predictors), EXPLORE is an attractive candidate if the aim is to develop an interpretable prediction model. Moreover, the optimal decision rules obtained by EXPLORE perform similar or better than GOSDT-G for all prediction tasks. Our findings indicate a trade-off between model performance and interpretability, with its strength varying across prediction tasks. In some cases, a more interpretable model can be found with minor performance loss. Most previous work examining the trade-off has used open-source benchmark datasets (e.g. OpenML) (17–19). However, routinely collected health care data is typically more complex as it contains many observations/features and is very sparse. Earlier work that did study health care data only considered a limited number of prediction tasks (at most two) and did not investigate many of the model algorithms in this paper (e.g. EXPLORE, IHT, GOSDT-G, RIPPER, XGBoost) (16, 20). We

further contribute to the existing empirical evidence by evaluating the performance in terms of AUROC instead of accuracy, which is more commonly used (15). AUROC is more suitable to evaluate clinical prediction models as accuracy is very sensitive to imbalanced data (i.e. low outcome rate) and the chosen risk threshold. To our knowledge, this is the largest study examining the interpretability-performance trade-off in EHR data.

5.2. Clinically optimal decision rules

Further, we showed that learning under constraints with EXPLORE can be used to find a set of near-optimal models. We found that the set of good candidate models is very large (1,381 – 20,320 for maximum rule length 5) for all prediction tasks. Moreover, we examined the Rashomon set of models and found there is a large variability in both the generalizability and features included. This shows there is a potential to find more clinically optimal decision rules. Earlier work concluded that a large Rashomon set means an equally well-performing, interpretable model is likely to exist and correlates with the existence of models with good generalizability (21, 47). The ability to enhance the interpretability of the models was demonstrated by the experiments incorporating domain knowledge; both incorporating age/sex and task-specific features resulted in decision rules with similar performance on average and very small differences overall. These constraints can also lead to significant reductions in computation time by reducing the search space. Finally, we showed we can improve the original $CHADS_2$ score with EXPLORE by allowing EXPLORE to select additional covariates on top of the original score. This approach can also be applied to model updating when there is a desire for model stability over time (i.e. similar features in the model), as it keeps the original model intact. Learning under constraints with EXPLORE provides a novel solution to open challenges for (optimal) rule-based methods such as scalability and addressing user preferences (48). Furthermore, access to the Rashomon set of models is an asset of EXPLORE, shared by only a few other model algorithms (e.g. TreeFARMS (49), FasterRisk (50)).

5.3. Limitations

In this study we have used model size as a proxy measure for interpretability, which might not directly translate to human interpretability. Likewise, we have not been able to assess the human-machine task performance, which is ultimately what matters in real-world applications. This may both impact the true interpretability-performance trade-off. To assess this, user studies need to be performed (51). Additionally, while we studied a wide range of prediction tasks, the trade-off and best choice of model algorithm may differ based on the specific clinical application, which was not investigated in this study.

5.4. Future work

First, we may continue to improve EXPLORE by further enhancements in speed thereby allowing longer decision rules (e.g. reducing the search space using reversed class labels) or supporting direct optimization over rank statistics (e.g. AUROC). Second, as external validation is a good indicator of model generalizability, this might expose more benefits of simple models and hence the interpretability-performance trade-off could be further investigated when transporting models to other databases. Finally, developing and testing search strategies to find the clinically optimal model from the Rashomon set of models is an interesting direction of research.

6. CONCLUSIONS

Our study shows that more complex models generally outperform simpler ones, confirming the expected trade-off between model performance and interpretability. However, the observed trade-off varies across prediction tasks. While none of the complex models systematically outperforms the others, the simple models show significant variations in performance. EXPLORE's ability to learn under constraints makes it an attractive candidate to find clinically optimal decision rules.

Declarations

Ethics approval and consent to participate

The current study was approved by the IPCI Governance Board under number 3/2023. No human participants were involved in the study and informed consent was not necessary.

Consent for publication

Not applicable.

Availability of data and materials

The datasets generated and/or analysed during the current study are not publicly available due to patient privacy and data protection concerns. The code for EXPLORE and the experiments is available on GitHub.

Competing interests

The authors declare that they have no competing interests.

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Author contributions

A.M., J.K. and P.R. conceived of the presented idea. A.M. designed the study, carried out the implementation, and performed the experiments. J.K, E.F., K.V., and P.R. provided critical feedback and helped shape the research. A.M. wrote the article with input from all other authors.

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Figures

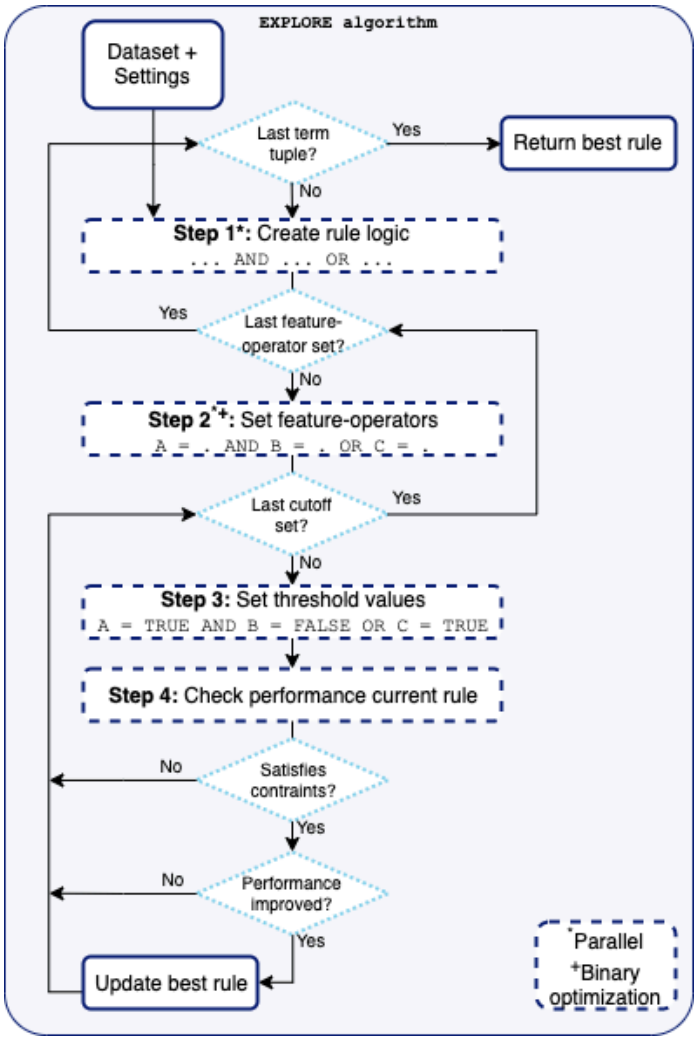

Figure 1

Overview of the improved EXPLORE algorithm.

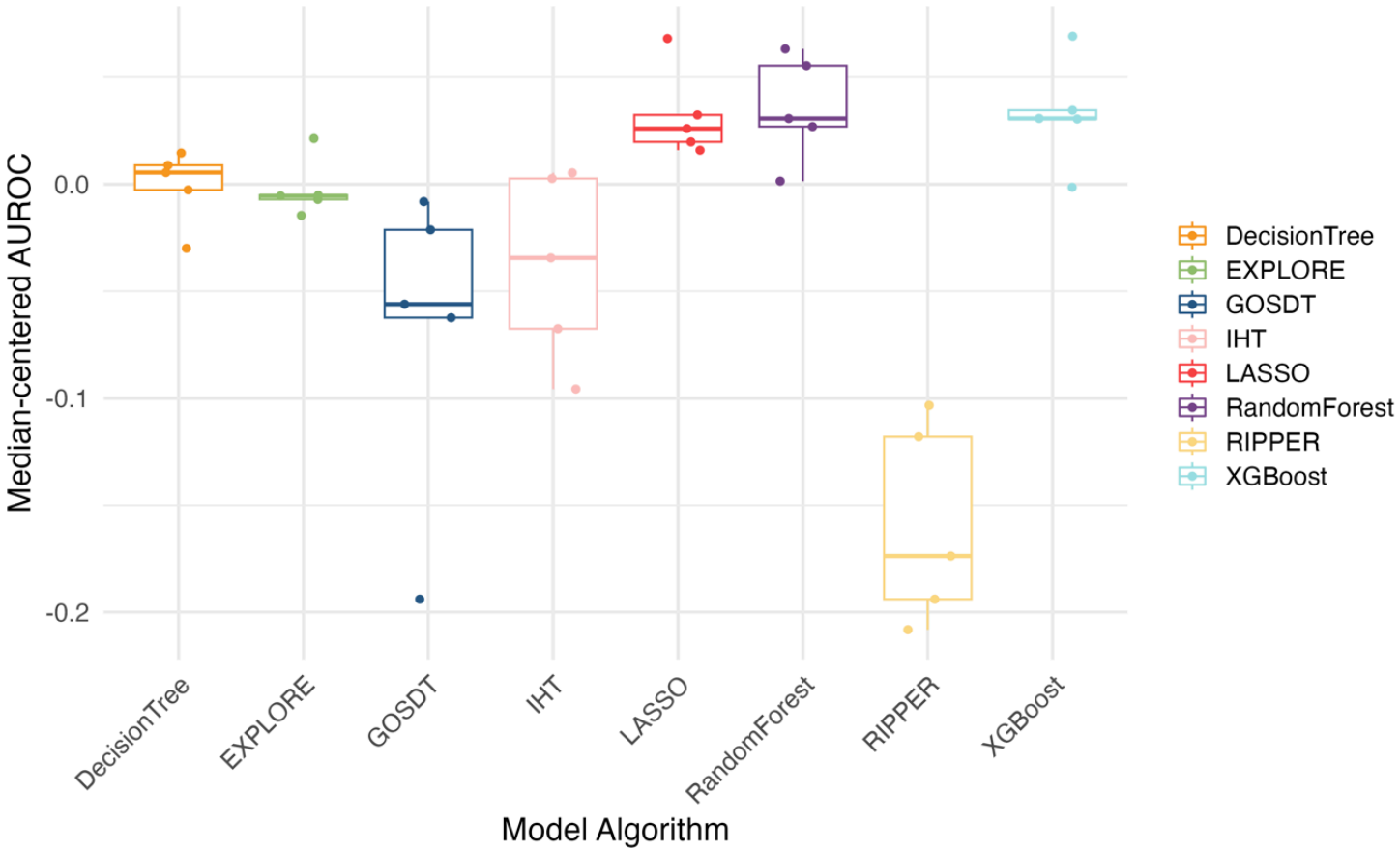

Figure 2

Median-centered AUROC on the test set averaged across five prediction tasks for different model algorithms.

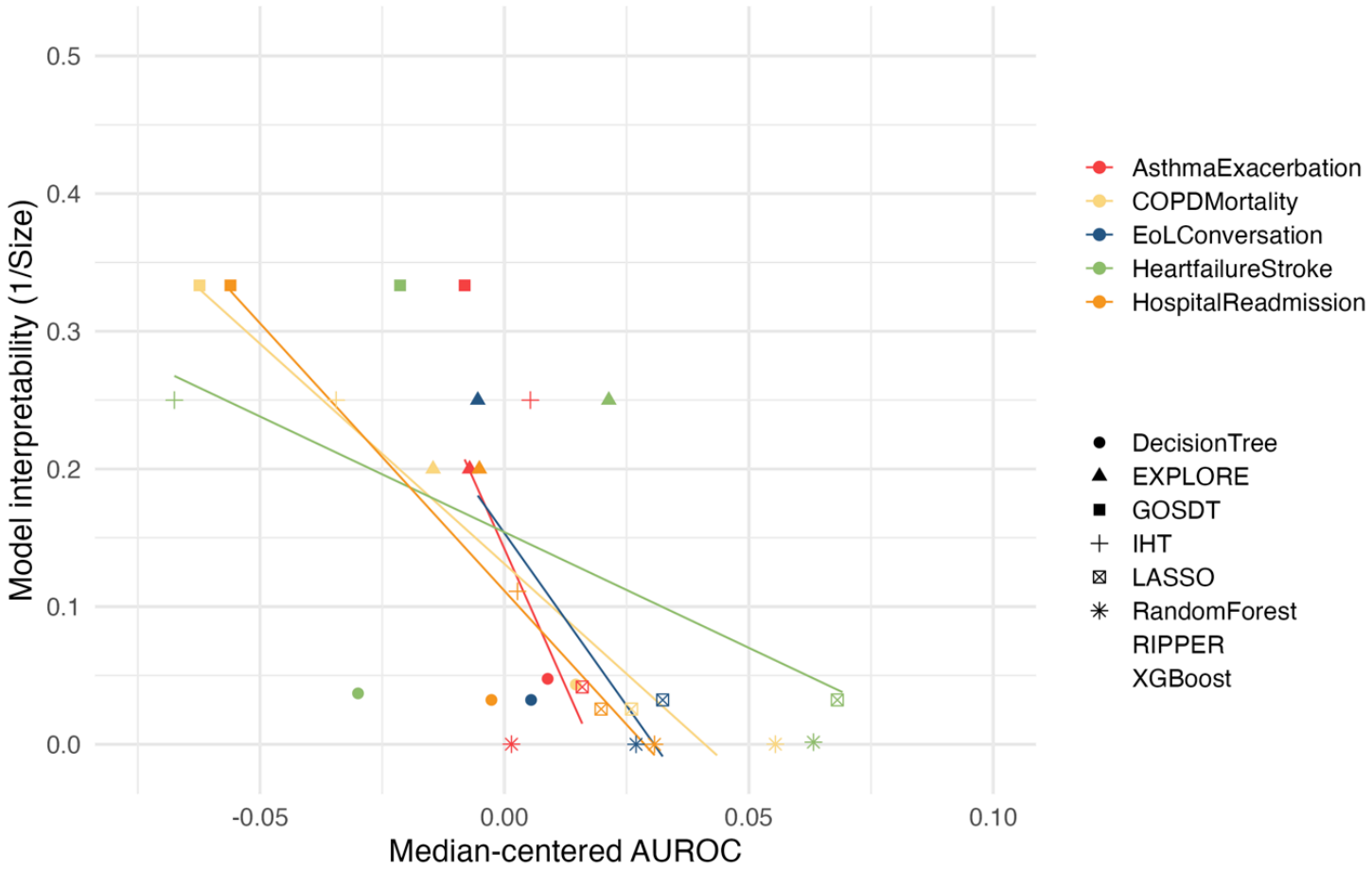

Figure 3

Model interpretability versus performance for prediction tasks on test set. Each dot corresponds to a prediction model; colors represent different model algorithms and shapes represent different prediction tasks. The lines (fitted using a linear model) indicate the observed relation between interpretability and performance across the different models for each prediction task.

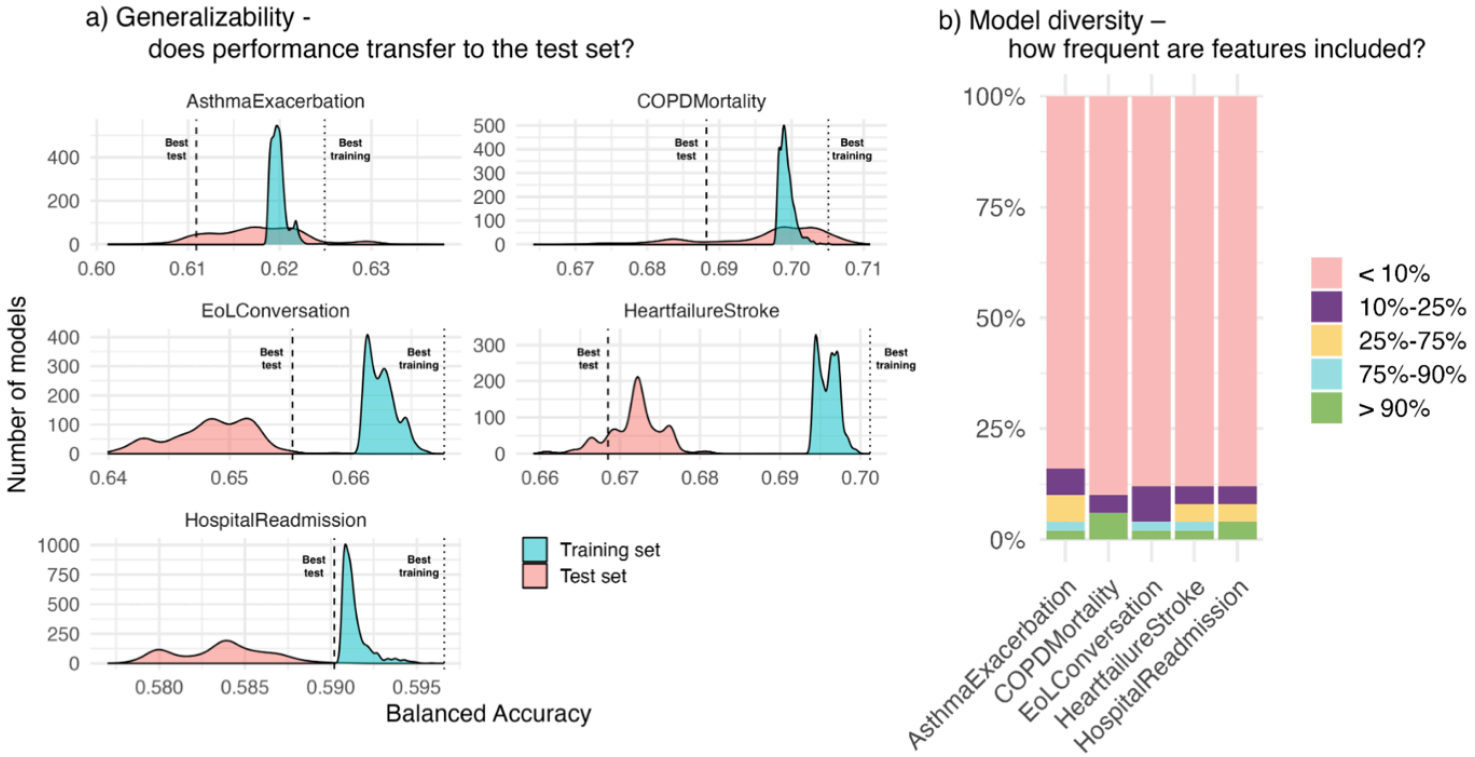

Figure 4

Characterizing EXPLORE's Rashomon set of models with less than 1% deviation from the balanced accuracy of the best model. The vertical lines indicate the performance of the best model in the training and test set.

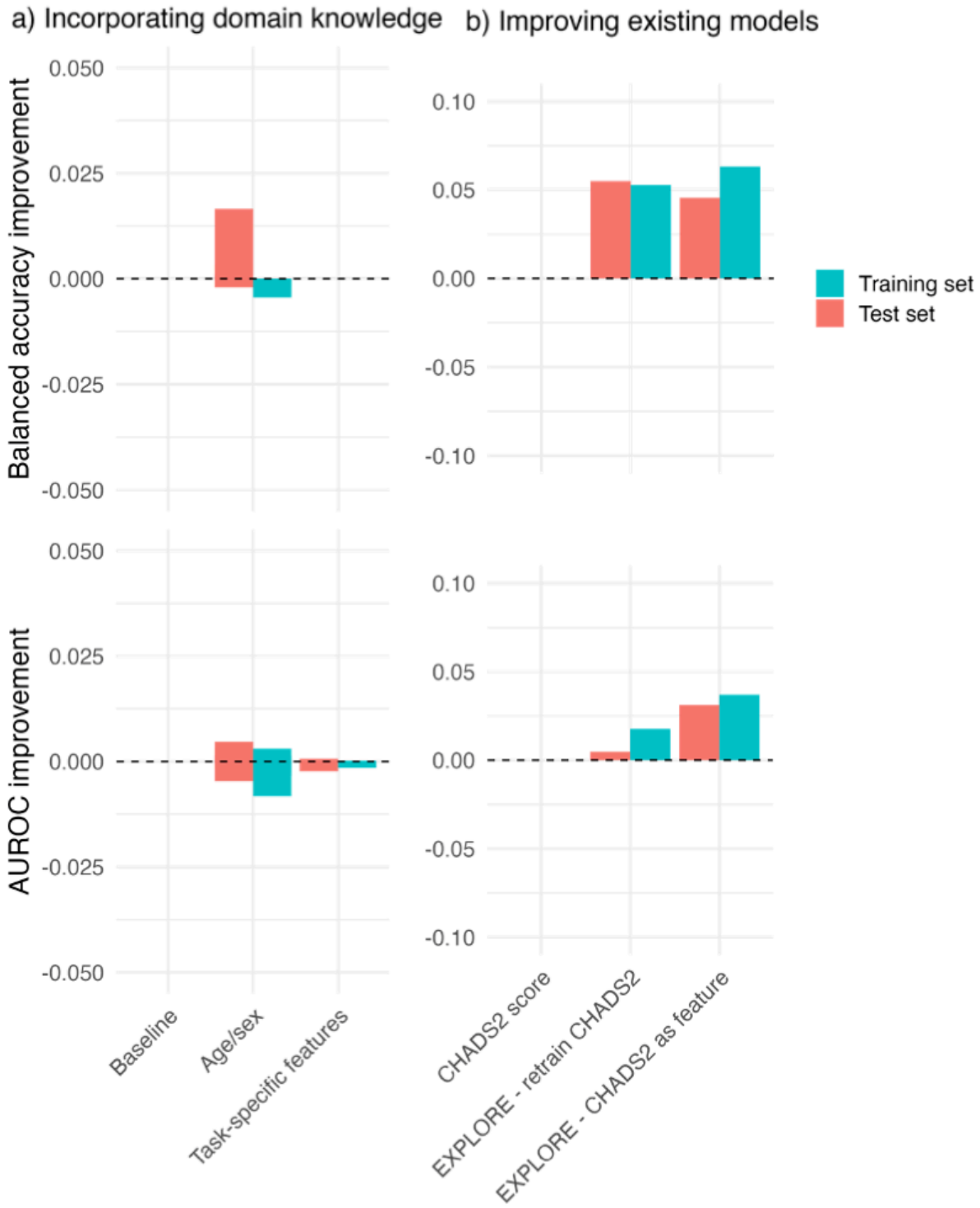

Figure 5

Improvement in AUROC and balanced accuracy values for mandatory features (a) and improved $CHADS_2$ score (b) with EXPLORE on training and test set. AUROC and balanced accuracy values can be found in Appendix D.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- EXPLORESupplementaryMaterial.docx